Review

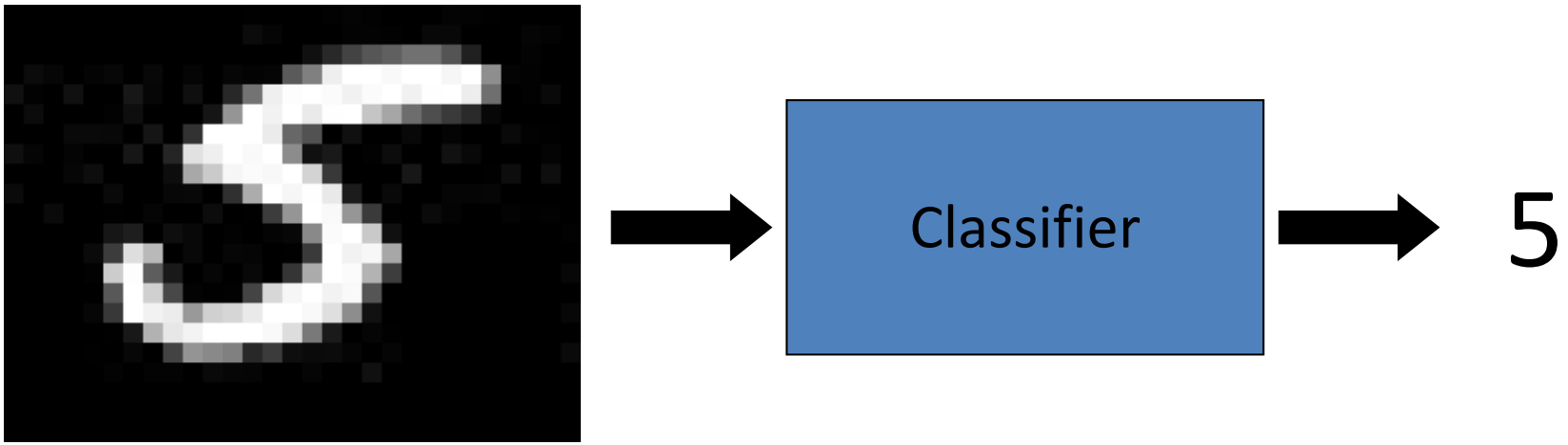
- Classification based on a single feature
- Two class classification
- Sample is assigned to one of the two classes
- The cost of making a false accept or a false reject is same

Bayesian Decision Theory

- Generalization of the preceding ideas
  - Use of more than one feature
  - Use more than two states of nature
  - Allowing actions other than decide on the state of nature
    - Allowing actions other than classification primarily allows the possibility of rejection
    - Refusing to make a decision in close or bad cases!
  - Introduce a loss function which is more general than the probability of error
    - The loss function states how costly each action taken is

Multiple Features

- **Digit Recognition**



- $X_1,\ldots,X_n \in \{0,1\}$ (Black vs. White pixels)
- $Y \in \{5,6\}$ (predict whether a digit is a 5 or a 6)

The Bayes Classifier

- Let's expand this for our digit recognition task:

$$P(Y=5|X_1,\ldots,X_n) = \frac{P(X_1,\ldots,X_n|Y=5)P(Y=5)}{P(X_1,\ldots,X_n|Y=5)P(Y=5)+P(X_1,\ldots,X_n|Y=6)P(Y=6)}$$

$$P(Y=6|X_1,\ldots,X_n) = \frac{P(X_1,\ldots,X_n|Y=6)P(Y=6)}{P(X_1,\ldots,X_n|Y=5)P(Y=5)+P(X_1,\ldots,X_n|Y=6)P(Y=6)}$$

- To classify, we will simply compute these two probabilities and predict based on which one is greater

Model Parameters

- For the Bayes classifier, we need to "learn" two functions, the likelihood and the prior
- How many parameters are required to specify the prior for our digit recognition example?
- How many parameters are required to specify the likelihood?
  - (Supposing that each image is 30x30 pixels)
- # of parameters for modeling $P(X_1,\ldots,X_n|Y)$:
  - $2(2^n-1)$

Model Parameters

- How many parameters are required to specify the likelihood?
  - (Supposing that each image is 30 x 30 pixels)

- # of parameters for modeling $P(X_1,\ldots,X_n|Y)$:
  - $2(2^n-1)$

- # of parameters for modeling $P(X_1|Y),\ldots,P(X_n|Y)$
  - $2n$



Model Parameters

- The problem with explicitly modeling $P(X_1,\ldots,X_n|Y)$ is that there are usually way too many parameters:
  - We'll run out of space
  - We'll run out of time
  - And we'll need tons of training data (which is usually not available)

$$\bar{\mathbf{x}} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i = \begin{bmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_j \\ \vdots \\ \bar{x}_K \end{bmatrix}$$

$$\mathrm{Cov}(A) = \begin{bmatrix} \frac{\sum(x_i-\bar{X})(x_i-\bar{X})}{N} & \frac{\sum(x_i-\bar{X})(y_i-\bar{Y})}{N} \\ \frac{\sum(x_i-\bar{X})(y_i-\bar{Y})}{N} & \frac{\sum(y_i-\bar{Y})(y_i-\bar{Y})}{N} \end{bmatrix}$$

$$= \begin{bmatrix} \mathrm{Cov}(X,X) & \mathrm{Cov}(Y,X) \\ \mathrm{Cov}(X,Y) & \mathrm{Cov}(Y,Y) \end{bmatrix}$$

The Naïve Bayes Model

- The *Naïve Bayes Assumption*: Assume that all features are independent **given the class label Y**

- Equationally speaking:

$$P(X_1, \ldots, X_n|Y) = \prod_{i=1}^{n} P(X_i|Y)$$

The Bayes Classifier

- A good strategy is to predict:

$$\arg\max_{Y} P(Y|X_1, \ldots, X_n)$$

  - (for example: what is the probability that the image represents a 5 given its pixels?)

- So … How do we compute that?

The Bayes Classifier

- Let's expand this for our digit recognition task:

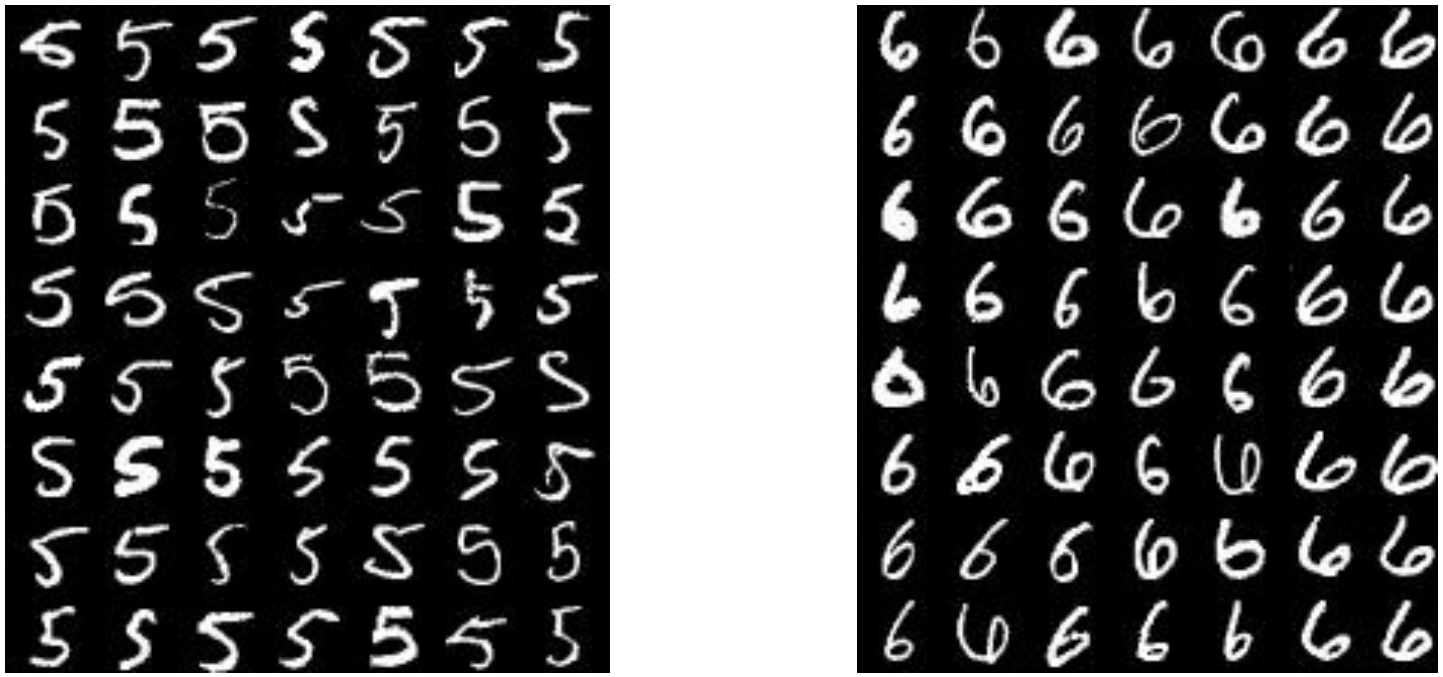
$$P(Y=5|X_1,\ldots,X_n) = \frac{P(X_1,\ldots,X_n|Y=5)P(Y=5)}{P(X_1,\ldots,X_n|Y=5)P(Y=5)+P(X_1,\ldots,X_n|Y=6)P(Y=6)}$$

$$P(Y=6|X_1,\ldots,X_n) = \frac{P(X_1,\ldots,X_n|Y=6)P(Y=6)}{P(X_1,\ldots,X_n|Y=5)P(Y=5)+P(X_1,\ldots,X_n|Y=6)P(Y=6)}$$

- To classify, we'll simply compute these two probabilities and predict based on which one is greater

Naïve Bayes Training

- Now that we've decided to use a Naïve Bayes classifier, we need to train it with some data:



MNIST Training Data

The Normal Density

- Univariate density
  - Continuous density
  - A lot of processes are asymptotically Gaussian
  - Handwritten characters, speech sounds are ideal or prototype corrupted by random process (central limit theorem)

$$N(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\,\sigma}\exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$

Univariate density

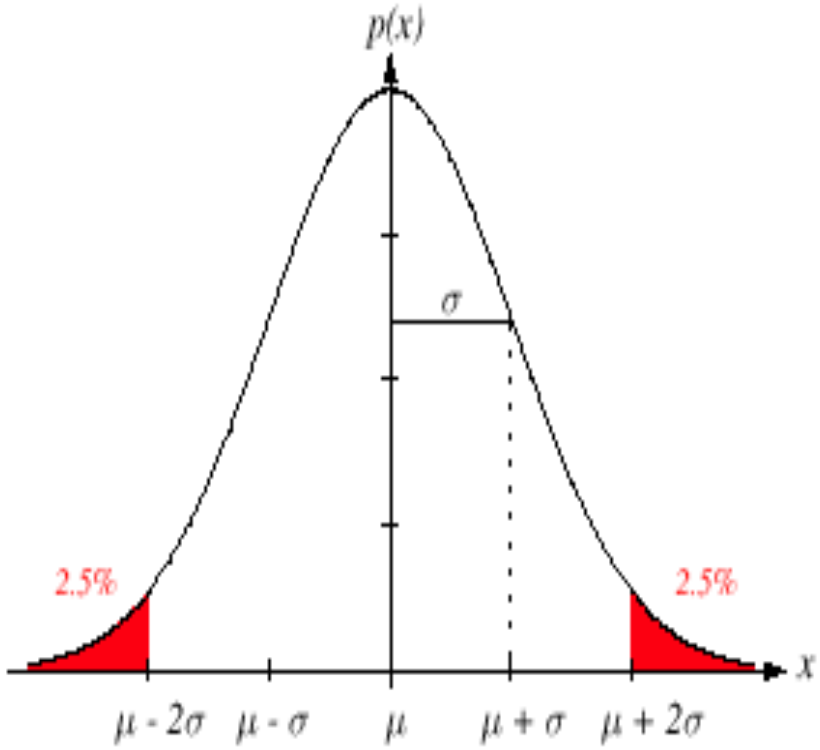


FIGURE 2.7. A univariate normal distribution has roughly 95% of its area in the range $|x - \mu| \leq 2\sigma$, as shown. The peak of the distribution has value $p(\mu) = 1/\sqrt{2\pi}\sigma$. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

The Normal Density

- Multivariate density: Multivariate normal density in d dimensions is:

$$N(x;\mu,\sigma^2) = \frac{1}{(2\pi)^{d/2}\left|\Sigma\right|^{1/2}}\exp\left[-\frac{1}{2}(x-\mu)^t\Sigma^{-1}(x-\mu)\right]$$

$x = (x_1, x_2, \ldots, x_d)^t$

$$cov_{x,y} = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{N-1}$$

$\mu = (\mu_1, \mu_2, \ldots, \mu_d)^t$ mean vector

$\Sigma = d*d$ covariance matrix

$|\Sigma|$ and Σ^{-1} are determinant and inverse respectively

Multivariate density

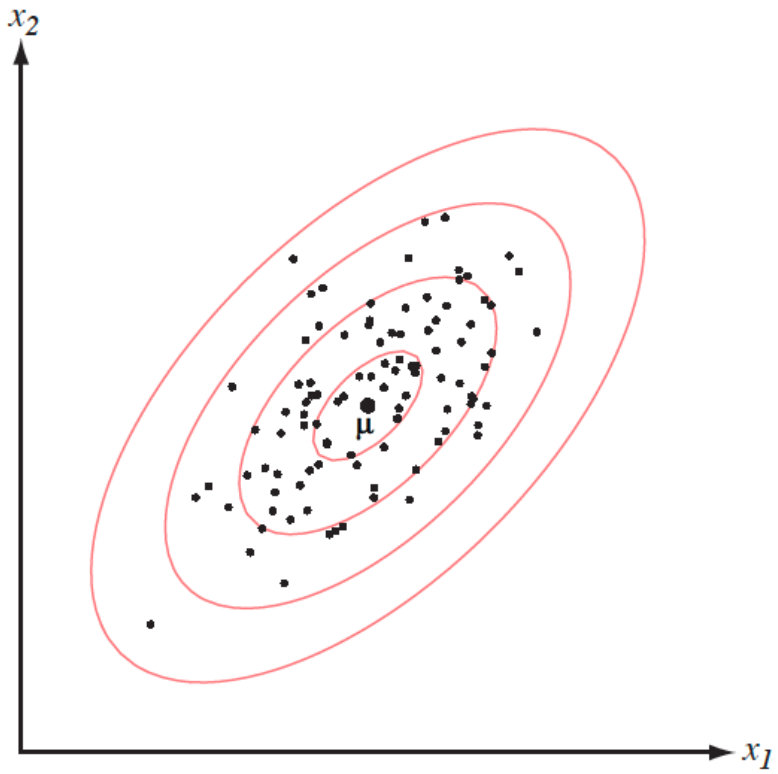


Figure 2.9: Samples drawn from a two-dimensional Gaussian lie in a cloud centered on the mean $\boldsymbol{\mu}$. The red ellipses show lines of equal probability density of the Gaussian.

Discriminant Functions for the Normal Density

$$P(\omega_j|x) = \frac{p(x|\omega_j)P(\omega_j)}{p(x)} \qquad P(error|x) = \min\left[P(\omega_1|x), P(\omega_2|x)\right]$$

- Minimum error-rate classification can be achieved by the discriminant function

- g1(x) = ln p(x | ω1) + ln P(ω1) – ln p(x)
- g2(x) = ln p(x | ω2) + ln P(ω2) – ln p(x)

Discriminant Functions for the Normal Density

$$P(\omega_j|x) = \frac{p(x|\omega_j)P(\omega_j)}{p(x)},$$

- Minimum error-rate classification can be achieved by the discriminant function
- gi(x) = ln P(x | ωi) + ln P(ωi)

- Case of multivariate normal

$$g_i(x) = -\frac{1}{2}(x-\mu_i)^t\Sigma_i^{-1}(x-\mu_i) - \frac{d}{2}\ln 2\pi - \frac{1}{2}\ln|\Sigma_i| + \ln P(\omega_i)$$

$$N(x;\mu,\sigma^2) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}}\exp\left[-\frac{1}{2}(x-\mu)^t\Sigma^{-1}(x-\mu)\right]$$

Analyzing Covariance Matrix

$$g_i(x) = -\frac{1}{2}(x-\mu_i)^t \Sigma_i^{-1}(x-\mu_i) - \frac{d}{2}\ln 2\pi - \frac{1}{2}\ln|\Sigma_i| + \ln P(\omega_i)$$

- Two cases:
  - Features are independent
  - Features are dependent

$\Sigma_i = \sigma_i{}^2.I$

Case $\Sigma_i = \sigma^2.I$ (I stands for the identity matrix)

Case $\Sigma_i = \Sigma$ (covariance of all classes are identical but arbitrary!)

Case Σi = actual covariance

Discriminant Functions for the Normal Density

- Case $\Sigma_i = \sigma^2.I$ (I stands for the identity matrix)
  - $\sigma_{ij} = 0$: Features are statistically independent
  - σ_{ii} is same for all the features

$$g_i(x) = -\frac{1}{2}(x-\mu_i)^t \Sigma_i^{-1}(x-\mu_i) - \frac{d}{2}\ln 2\pi - \frac{1}{2}\ln|\Sigma_i| + \ln P(\omega_i)$$

$1/\sigma^2$

Constant for all the classes

Constant for all the classes

Discriminant Functions for the Normal Density

- Case $\Sigma_i = \sigma^2.I$ (I stands for the identity matrix)
  - $\sigma_{ij} = 0$: Features are statistically independent)
  - σ_{ii} is same for all the features

$$g_i(x) = -\frac{1}{2}(x-\mu_i)^t\Sigma_i^{-1}(x-\mu_i) - \frac{d}{2}\ln 2\pi - \frac{1}{2}\ln|\Sigma_i| + \ln P(\omega_i)$$

$$g_i(\mathbf{x}) = -\frac{1}{2\sigma^2}[\mathbf{x}^t\mathbf{x} - 2\mu_i^t\mathbf{x} + \mu_i^t\mu_i] + \ln P(\omega_i)$$

Discriminant Functions for the Normal Density…

$$g_i(\mathbf{x}) = -\frac{1}{2\sigma^2}\left[\mathbf{x}^t\mathbf{x} - 2\mu_i^t\mathbf{x} + \mu_i^t\mu_i\right] + \ln P(\omega_i)$$

- Disregarding $x^t x$, we get a linear discriminant function

$$g_i(x) = w_i^t x + w_{i0}$$

where :

$$w_i = \frac{\mu_i}{\sigma^2};\;\; w_{i0} = -\frac{1}{2\sigma^2}\mu_i^t\mu_i + \ln P(\omega_i)$$

(ω_{i0} is called the threshold for the ***i***th category!)

Discriminant Functions for the Normal Density…

- A classifier that uses linear discriminant functions is called “a linear machine”
- The decision surfaces for a linear machine are hyperplanes defined by $g_i(x) = g_j(x)$

$$\mathbf{w}^t(\mathbf{x} - \mathbf{x}_0) = 0 \qquad \mathbf{w} = \boldsymbol{\mu}_i - \boldsymbol{\mu}_j$$

$$\mathbf{x}_0 = \frac{1}{2}(\boldsymbol{\mu}_i + \boldsymbol{\mu}_j) - \frac{\sigma^2}{\|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|^2} \ln \frac{P(\omega_i)}{P(\omega_j)} (\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)$$

Discriminant Functions for the Normal Density…

– The hyperplane separating $\mathcal{R}_i$ and $\mathcal{R}_j$

$$x_0 = \frac{1}{2}(\mu_i + \mu_j) - \frac{\sigma^2}{\|\mu_i - \mu_j\|^2} \ln \frac{P(\omega_i)}{P(\omega_j)} (\mu_i - \mu_j)$$

is always orthogonal to the line linking the means!

$$if \;\; P(\omega_i) = P(\omega_j) \;\; \text{then} \;\; x_0 = \frac{1}{2}(\mu_i + \mu_j)$$

$$g_i(x) = -\|x - \mu_i\|^2$$

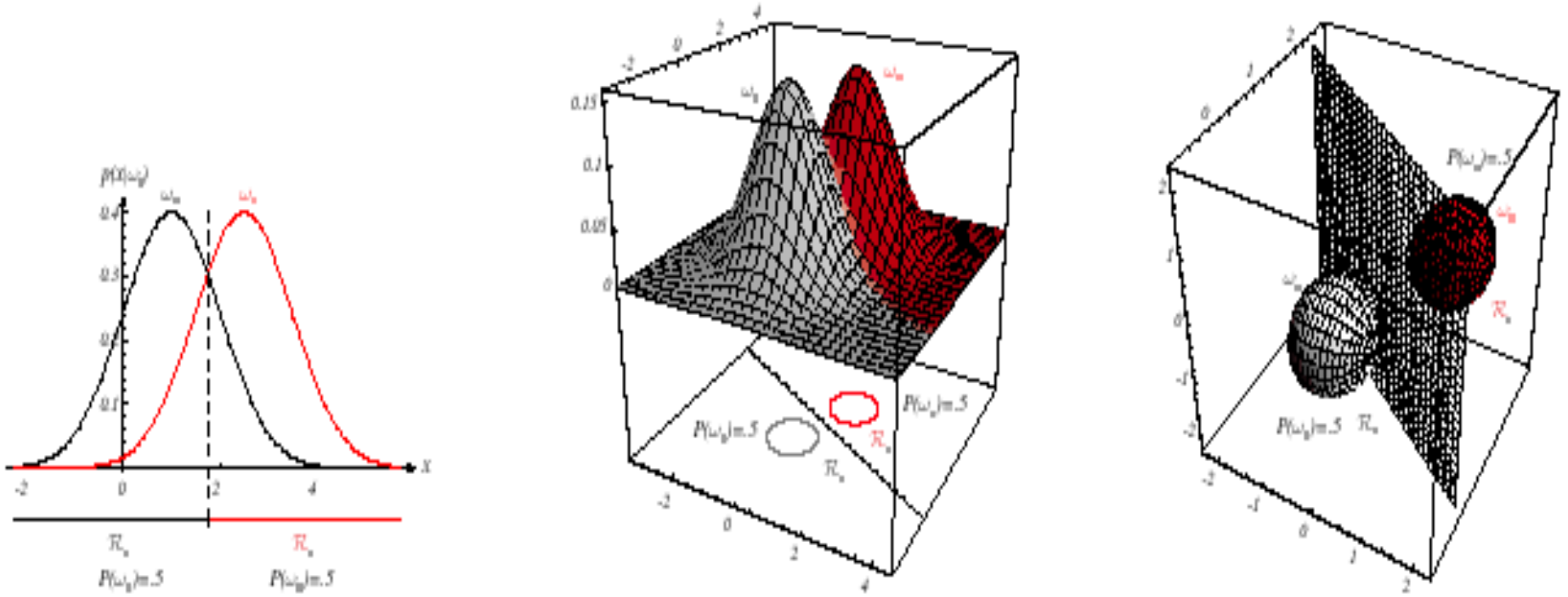


FIGURE 2.10. If the covariance matrices for two distributions are equal and proportional to the identity matrix, then the distributions are spherical in d dimensions, and the boundary is a generalized hyperplane of $d-1$ dimensions, perpendicular to the line separating the means. In these one-, two-, and three-dimensional examples, we indicate $p(\mathbf{x}|\omega_i)$ and the boundaries for the case $P(\omega_1) = P(\omega_2)$. In the three-dimensional case, the grid plane separates $\mathcal{R}_1$ from $\mathcal{R}_2$. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

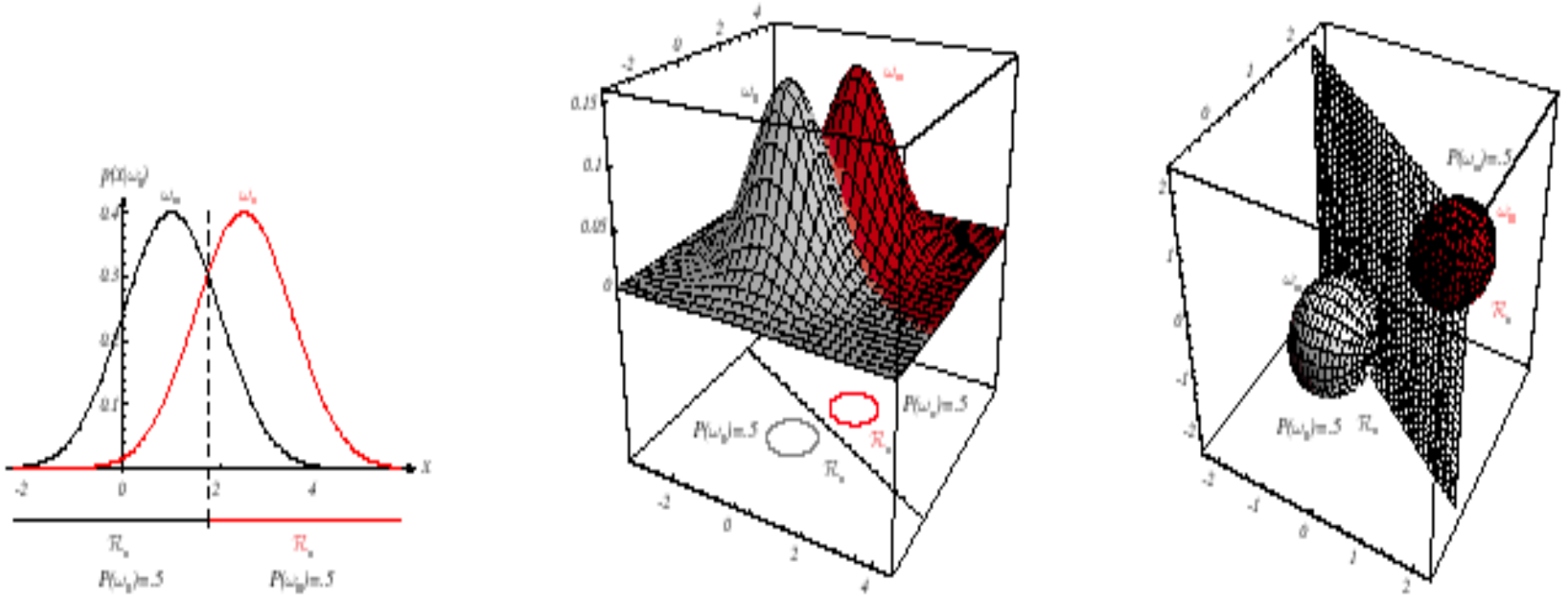
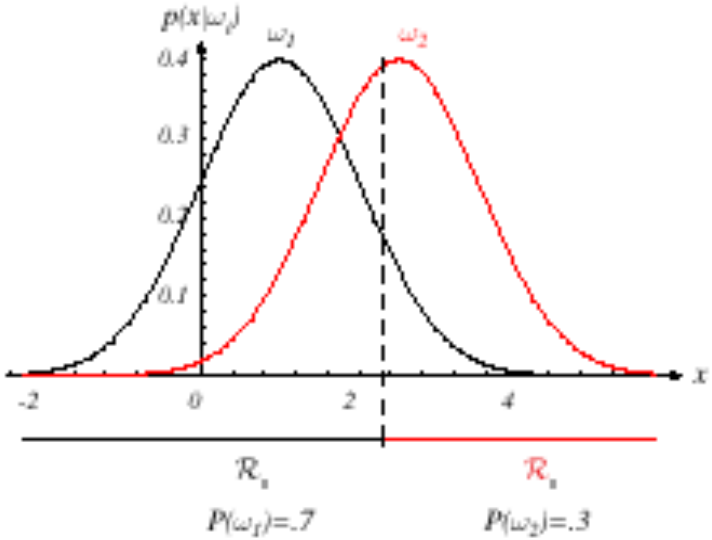
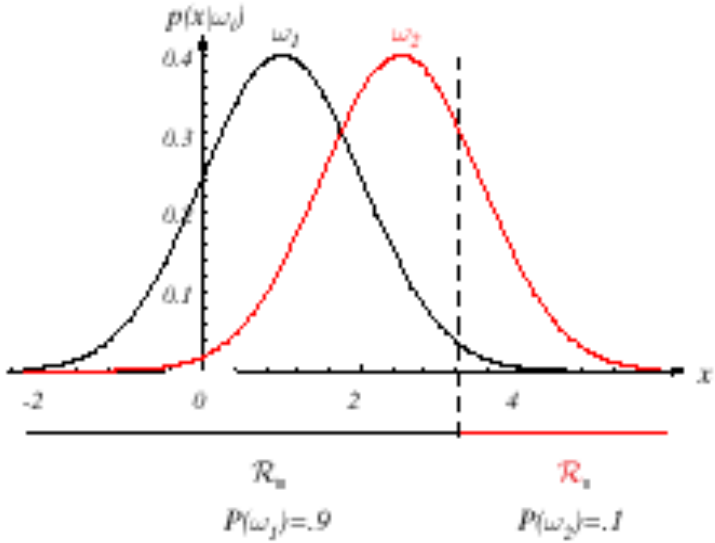
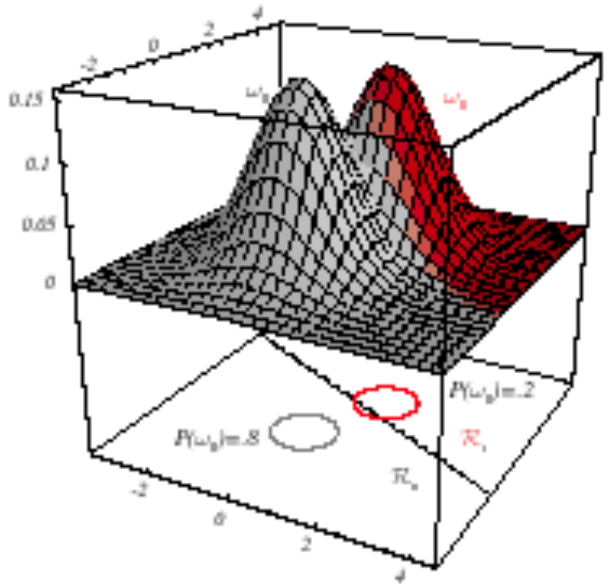
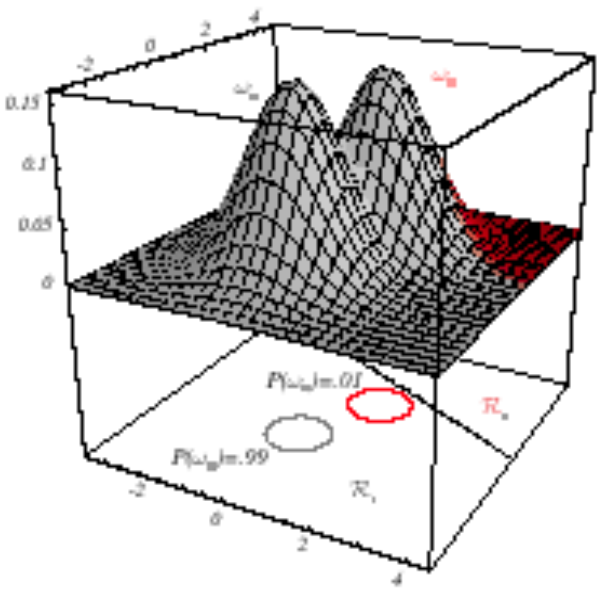


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p(x|ωj)
ω1
ω2
0.4
0.3
0.2
0.1
x
-2
0
2
4
R1
R2
P(ω1)=.7
P(ω2)=.3


p(x|ωj)
ω1
ω2
0.4
0.3
0.2
0.1
x
-2
0
2
4
R1
R2
P(ω1)=.9
P(ω2)=.1


0.15
0.1
0.05
0
P(ω2)=.2
P(ω1)=.8
R2
R1


0.15
0.1
0.05
0
P(ω2)=.01
P(ω1)=.99
R2
R1

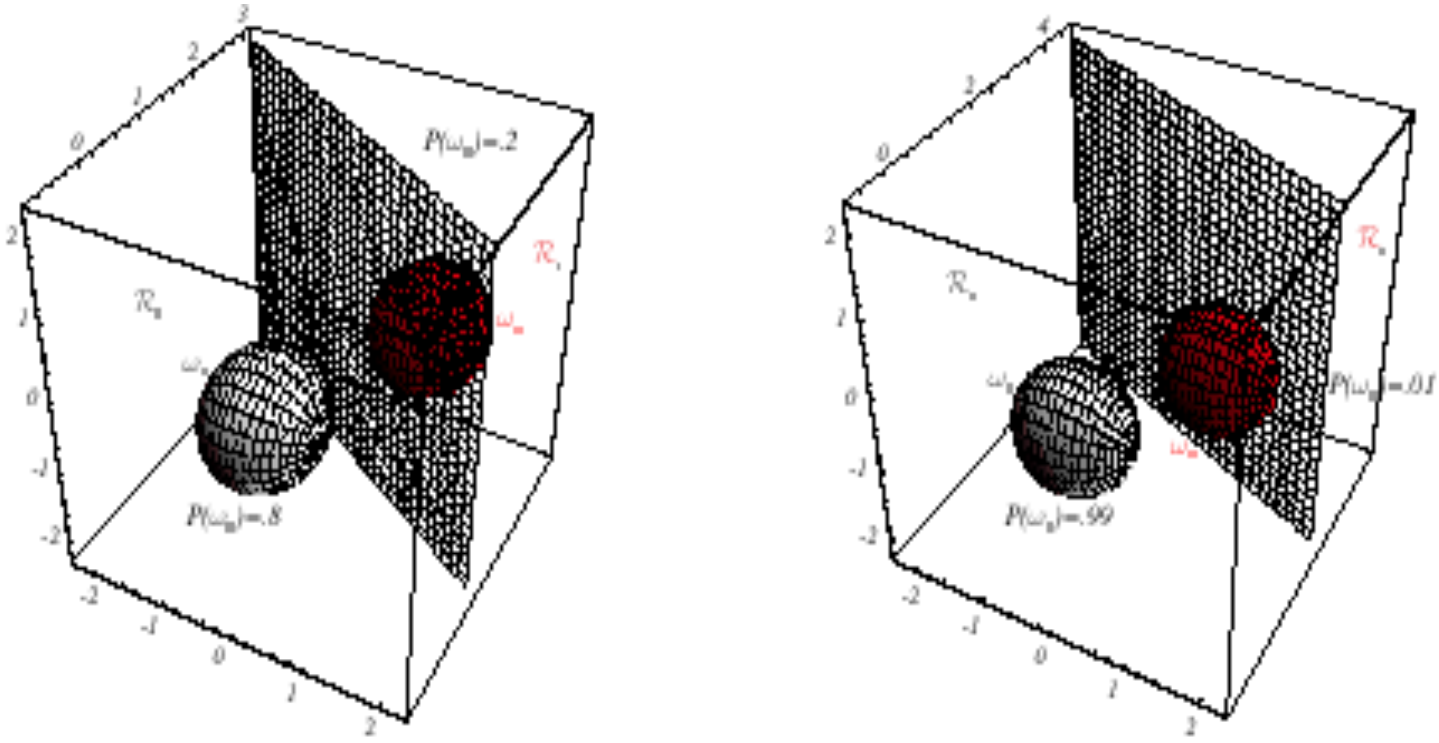


FIGURE 2.11. As the priors are changed, the decision boundary shifts; for sufficiently disparate priors the boundary will not lie between the means of these one-, two- and three-dimensional spherical Gaussian distributions. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

The weather data, with counts and probabilities												
outlook			temperature			humidity			windy		play	
	yes	no		yes	no		yes	no		yes	no	yes	no
sunny	2	3	hot	2	2	high	3	4	false	6	2	9	5
overcast	4	0	mild	4	2	normal	6	1	true	3	3		
rainy	3	2	cool	3	1								
sunny	2/9	3/5	hot	2/9	2/5	high	3/9	4/5	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	mild	4/9	2/5	normal	6/9	1/5	true	3/9	3/5		
rainy	3/9	2/5	cool	3/9	1/5								

A new day				
outlook	temperature	humidity	windy	play
sunny	cool	high	true	?

The weather data, with counts and probabilities												
outlook			temperature			humidity			windy			play	
	yes	no		yes	no		yes	no		yes	no	yes	no
sunny	2	3	hot	2	2	high	3	4	false	6	2	9	5
overcast	4	0	mild	4	2	normal	6	1	true	3	3		
rainy	3	2	cool	3	1								
sunny	2/9	3/5	hot	2/9	2/5	high	3/9	4/5	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	mild	4/9	2/5	normal	6/9	1/5	true	3/9	3/5		
rainy	3/9	2/5	cool	3/9	1/5								

A new day				
outlook	temperature	humidity	windy	play
sunny	cool	high	true	?

- Likelihood of yes

$$= \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14} = 0.0053$$

- Likelihood of no

$$= \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{14} = 0.0206$$

- Therefore, the prediction is No

- Likelihood of yes

$$= \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14} = 0.0053$$

- Likelihood of no

$$= \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{14} = 0.0206$$

- Therefore, the prediction is No

Assume $p(x|\omega_1) \sim N(5,1)$ and $p(x|\omega_2) \sim N(6,1)$. Without performing any explicit calculations, determine x^* for the minimax criterion. Explain your reasoning.

$$\mathbf{x_0} = \frac{1}{2}(\boldsymbol{\mu}_i + \boldsymbol{\mu}_j) - \frac{\sigma^2}{\|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|^2} \ln \frac{P(\omega_i)}{P(\omega_j)} (\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)$$

Assume $p(x|\omega_1) \sim N(0,1)$ and $p(x|\omega_2) \sim N(1/2,1/4)$, under a zero-one loss. Find x^* and the overall minimax loss.